8 User Representations: Practices, Methods and Sociology

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Technological developments impinge on a society composed of complex and many-sided users moving within a heterogeneous set of relationships. Users are simultaneously consumers, citizens, members of a family, of a religious faith, of a social club, of a political party, of a profession, or whatever. In many cases, technical design debates turn on the need to arbitrate between the varying positions of the user which are more or less incompatible within a single technological system. Accordingly creating a market invariably means tuning in to several wavelengths simultaneously. The way an innovation takes hold cannot be reduced to any one model, e.g. the rational economic consumer, the status-seeking purchaser or the inveterate trend-follower. Instead, the success or failure of innovations frequently depends on their ability to cope with dissimilar users possessing widely differing skills and aspirations. The micro computer market is a very instructive example here. The importance of users in technological development has been recognized by a number of economists. In particular, they have shown the following:

1) The process of innovation does not end when a product is launched. The final destiny of an innovation depends largely on the recruitment of first users (David 1986a,b) and on their ability to generate a whole series of improvements affecting not only the proposed technology but also elements of its environment and/or complementary technologies (Rosenberg 1976 and Teece 1986).
2) In certain technological areas users play a predominant role in the conception and diffusion of an innovation while entrepreneurs mainly take care of producing the device (Von Hippel 1976 and 1988).
3) The development and commercialization of innovations are based on interactions between the innovators and their potential users, a process which enables both sides to improve their knowledge and which allows for small adjustments to the products (Andersen and Lundvall 1988; Meyers and Althaide 1991).

In this chapter I will extend this economic analysis of users in technological development. Economists mainly focus on how the interaction between innovators and certain users allows for adjustment between an offer that already exists and a demand which will only define itself during the interaction. I go one step further and argue that innovators are from the very start constantly interested in their future users. They construct many different representations of these users, and objectify these representations in technical choices.

The different forms of user representation generated during innovation processes are of critical importance for constructive technology assessment (CTA) which aims at modulating technical change. CTA must seek to discover how it is possible, in the processes of developing and marketing a technological system, to reconcile what may be barely compatible and even conflicting user representations. For example, many opinion polls show that a majority is in favour of reducing the country's energy consumption (whether on the grounds of economy, of environmental conservation, of national self-sufficiency, or whatever). A mere appeal, however, to people's sense of responsibility is not enough to produce a new type of energy-saving user. What is needed are application systems superimposing the user representations of personal comfort (entailing high energy consumption) with careful consumption of energy. Realizing this superposition requires the possession of representations (in a cognitive and political sense) of what the supposed users are and what they want (Compare Downey, this volume).

In this chapter, I shall describe the variety of techniques employed by system designers to construct and then to appropriate user representations. This analysis will make clear that the challenge for CTA is not so much devising new user representations, for there are many of them already. CTA must discover methods for coping with the many existing representations, so that they ultimately converge in a way combining different representations and enabling the initial development programme to go forward. Going back to our energy-saving example, questions need to be answered. First, how to bring users who may be insensitive to national policy demands and beset by a variety of more immediate problems and constraints, to take an interest in energy-saving systems? Second, once energy-saving systems have been adopted, how to ensure their use will in fact save energy? This is important, because users may always deform or defeat the initial purpose of a technological system (see Akrich 1992 as well).

Throughout the chapter three case studies will be discussed.[1] All three are standard technological systems for the mass market:

- an interface terminal linking a TV cable network to its subscribers, known as the CA unit ("coffret d'abonné") and basically designed for TV programme selecting;
- a range of telephones known as the "Contact Ambiance" range and designed for household use;
- a domestic computerized system known as "Securiscan," which has a number of applications ranging from house surveillance to programmed heating and remote-control of automatic devices, adaptable according to user specifications.

The CA unit was supported by the French Telecom administration and developed in close collaboration with the manufacturers, whereas the other two systems are entirely private developments. These three systems illustrate that the creation of successful artifacts depends on the ability of innovators to generate user representations and integrate them into their designs, and that alternative strategies are available for aligning the various user positions. When analysing these strategies, I shall extend the scope of the innovation process beyond its generally accepted boundaries, to encompass marketing and policy decisions that illuminate lines of thought and action relevant to a constructive technology assessment.

User Representation Techniques

In this section I describe all the techniques for producing user representations observed in our research. Our approach diverges fundamentally from that of the market researcher who deals with only a fraction of the full range of methods deployed in connection with designing and developing technological systems. We deliberately refrain from making a prima facie distinction between techniques "legitimized" by a formal scientific and conceptual basis, and those of a more empirical kind lacking any such basis. In our approach, we consider that the mere fact that an argument introduced into the project discussions was put forward in the name of the user makes it relevant to our purposes. This brings us to classify the techniques observed in two categories: *explicit techniques*, based on special skills or qualifications in the area of defining or interpreting consumer representations; and *implicit techniques*, which rely on statements made on behalf of the users.

Explicit Techniques

Market Surveys

Market surveys enjoy a very special status. When questioned about users, the authors of an innovation immediately quote the market

survey as a pillar supporting their perception of the future users of their systems. When pressed for specifics, however, they become more evasive and produce nothing more than a few broad generalizations hinting at the potential strong demand for products of a certain kind. Researchers who persist by trying to obtain a copy of the survey report are likely to be disappointed: in only one of the three cases investigated were we able to find any written trace of such a study. Our experience is that this strange state of affairs reflects that, in many cases, these studies are used to persuade the higher decision-making authorities to support the project, but that these studies are rarely consulted during the subsequent development phase. In other words, actual practice is based on a division of responsibilities between the technical and the marketing sides, the former handing over a complete product package that the latter are expected to sell "as found." In these conditions, the market survey may be used to identify buyers, or to put the final touches on the sales argument, but it certainly has nothing to do with the design and development of the actual product.

The one market survey report we obtained, for the "Contact Ambiance" telephone, confirmed this interpretation. It was issued when the product design phase was nearly completed. Its salient indicators were of the socio-economic type ("originality" of outward appearance and price). The report did not include technical recommendations, thus leaving every discretion to the designers. The purpose of the market survey was not to reveal opportunities for creating a new demand, but (assuming an existing demand) to identify anything which might restrain the buying impulse.

Consumer Testing

When a product emerges from its design cocoon, its designers often organize tests directed to a sample group deemed to be representative of its future users. The nature and coverage of such tests can vary a great deal and, in the case of the CA terminal, two different types were used. The first type of test was conducted with personnel of the sponsoring administrations and the manufacturer. The aim was to find out which of several different presentations being considered for the unit would meet with the widest approval. Other questions related to the place reserved for the unit within the "domestic layout arrangement." The survey was of the "democratic" type assuming the existence of an average standard of taste, minimizing the number of dissatisfied users, and weighting heavily the user motivation factor.

The consumer test of the "Contact Ambiance" telephone consisted of introducing strongly contrasting outer presentations of technically

identical products. This was in direct contradiction with the prior assumptions underlying the test described above. Thus, it is obscure why the test was done in the first place. The markets targeted by the other two products were entirely dissimilar. For instance, it was not the interface terminal (CA unit) itself which was marketed, but the whole system carried by the video communications network. This case makes clear that the nature of a test depends on the kind of market that is expected. In other words, the implementation of a particular consumer testing technique amounts to making one or more assumptions about *which* market is expected to expand.

The second type of test was contracted out to specialized institutes independent of the project sponsors. The purpose was to confirm the CA unit's ergonomic validity. At the same time, the interviewees were asked to give opinions concerning the CA unit, its place within the household, and the services to which it gave access. Here, the user was a "system operative" approached through a "school exam" test. The way the questions were formulated, presented and contextualized, situated the examinee as a user of the technological system. Tests of this type cannot be expected to cover, once and for all, the whole range of problems likely to face the user in contact with the system. Generally speaking, the "classroom" situation places special emphasis on some problems, whereas some others arising only when the network is in full-scale operation go unnoticed.

One set of unnoticed problems evolved around security procedures. In the proposed CA unit, access to services subject to surcharges was protected by a secret code word which had to be keyed in whenever these channels were called up. This enabled the subscriber to control access to these surcharged services, e.g., by preventing children in the house from running up unwanted bills. Once the user selects a restricted-access channel, if he presses the "delete" key before entering his code word in full, he is returned to the general-access channel and has to recommence the entire procedure including selection of the restricted channel. Here, the system assumes that the user has abandoned his intention to call up the service in question, either because he does not wish to incur the charge or because he finds himself ineligible (not possessing the code word). This situation is not foreseen in the test scenarios designed to verify understanding of operations instructions and procedures. The test assumes that, in all cases, the user knows what he wants and is duly entitled to obtain it. The test does stimulate a rather different situation, however, namely that the user has committed an error when entering his access code, but still wishes to call up the restricted channel by re-entering the code correctly. In this situation,

being returned to the general access channel is found either a nuisance or an inexplicable aberration.

Thus, although such tests may often have the outward appearance of an encounter between humans and machines, we must not forget that they do not cover the full extent of the relationships between the two. They rely on "typical" and simplified situations for the purposes of demonstrating particular user responses. This makes them efficient, but at the same time limits their true relevance.

Feedback On Experience

Through contacts with after-sales services, equipment suppliers, and other customer services, the "real" users feed back information to technological system designers. But this information is twice filtered: by the users themselves, who pass on only those remarks they consider relevant to their relationship with the system itself or with its agents; and by the latter, for similar reasons. In the cases we studied, this feedback was one of the factors causing modifications. For example, it was only in real-life situations that a problem with the aiming of the CA unit remote-control pad was revealed. The pad was effective only if pointed directly at the unit itself. The test conditions, although designed to challenge the competence of prospective users, totally failed to detect problems of this kind. Subjects of testing are much more concentrated on this particular task than users, sitting comfortably at home in front of their TV screens. The "laboratory" situation imposed by the test protocol was unrealistically strict. The real-life user refuses to be represented as a dedicated keyboard operator. While using the remote-control pad he wants to be able to maintain normal relationships with his environment: to take part in a conversation, to glance at the programme schedule, to help himself to a drink or a bit of food, and so on.

In the case of the "Contact Ambiance" telephone, the after-sales department insisted on adding a bracket enabling the user to put down the handset, without hanging up, which is needed, for instance, when he has to go and look for information requested by his correspondent. The designers had represented and met consumer-purchaser wishes such as easy storing, by providing a wall-mounted telephone (not a usual option in the business world to which the manufacturer has hitherto been accustomed). But they had not simulated a comprehensive user universe, which might have revealed the unfortunate spectacle offered by a telephone dangling forlornly in space. Although, as our case studies show, feedback of after-sales agencies can suggest design changes, it is not always possible to modify the system to meet every

complaint or suggestion. In some cases this could create serious difficulties on the production and organizational sides.

Implicit Techniques

Market surveys and testing have the common aim of seeking to simulate the prospective user's behaviour and reaction. They present "typical" scenarios acted out by people deemed to be representative of real-life users. However, they are not the only producers of user representations present in the design-development processes. Feedback via after-sales services provides much valuable additional input. Other less-formal techniques can be identified as well. These techniques have in common that they actually address the "real" users, but rely on spokespersons of three general types: designers, expert consultants, and other products.

"I..."

Reliance on personal experience, whereby the designer replaces his professional hat by that of the layman, is a much more common device than might be thought at first sight. When there is no other available means of bringing in the end-user, or when organized test procedures seem too complicated or too expensive, arguments uttered from the founts of inspiration can carry a certain amount of conviction. In any event, reliance on personal experience was used throughout the design and development phase of the CA unit. This was no doubt due to some degree of isolation of the design team. It was heavily dominated by engineering specialists, with nobody to represent the marketing side, no true ergonomics expert (even if some engineers thought they could handle this aspect), no media representative, and no operator of a comparable service.

The Experts

The user is brought into the equation without ceremony but with great impact depending on the way that a company or project team organizes the development of new products. The process is initiated by working procedures which enable effective collaboration between the technical and the marketing departments although, as we found in the "Contact Ambiance" study, this does not always occur. In that case, the marketing side won authoritative status only after a hard struggle. It is not so much that marketing people are more authorized to represent the users than their technical counterparts. In fact, giving too much

authority to marketing people who by definition are more concerned with the user's market role (purchaser, consumer etc.) may result in neglecting the user in his technical role (machine operator and "fixer" etc.). In our cases introducing this particularly technical facet of the user occurred only in a late stage of product development, i.e. at a point where it was not easy to revise the technological options. These findings show that "the user" is not a single entity taken on board when the project is launched, but a set of disparate characteristics which will not necessarily merge into a tight configuration ready to accommodate the definitive end-user. The presence in the project organization of a wider range of disciplines, when they all contribute to formulating the technical options, may expedite and facilitate the amalgamation of these disparate characteristics.

The production of the user's manual for the CA unit was an illustration of how user representation of experts can be brought into the project organization. A special group composed of new kinds of specialists, separate from the design team proper, was set up to write the instruction manual. In particular, the new group included a specialist who had been connected with the first French experiment with optical-fibre cable networks (Biarritz pilot experiment) and another who had contributed to the user's manual for the Minitel system. Thus, the project organization "co-opted" experts to produce user representations on the basis of their experience of user relations in similar technological systems.

Other products

Yet another option to elicit user representations is to adopt or reject representations present in products considered to have something in common with the innovation project in hand. This assumes the existence of relationships between the technical options under consideration and the type of users of the equivalent product. The adoption or rejection of an existing formula is in itself a more or less explicit decision concerning the type of user being targeted and his/her tastes, expectations, etc. Making such a decision requires calling up the particular representation of the user incorporated in the comparable product. We found examples of this in all three of our case studies.

From the outset the Securiscan project, a system for controlling all kinds of domestic applications, was defined in *opposition* to existing systems. The aim was to develop a "wire-free" installation, whereas all the alternatives involved wiring networks.[2] This approach was based on the marketing concept that additional wiring constituted, on both financial and aesthetic grounds, a major buying constraint. It was

argued that the market would expand significantly once this constraint had been removed.

The manufacturer of the "Contact Ambiance" telephones referred back to options for products he had already supplied to business users, sometimes incorporating the options and sometimes discarding them. For example, amplified reception had become a standard feature of business installations and it was retained for the top-of-the-line domestic model. This decision sharply contrasted with the general trend to omit amplification at that end of the domestic market. However, for reasons not made entirely clear, the sound amplification needed to be of higher quality for domestic than for business purposes. Finally, the "confidential" function, also a very common feature of business installations (it cuts off the microphone) was omitted for the domestic models on the grounds, assumed specific to this market, that it fulfilled no useful purpose since users did not trust it and continued to cup their hands over the microphone.

Finally, the CA unit keyboard as well as the user-CA dialogue was designed on the same lines as that of the Minitel terminal, as opposed to that of TV remote-control pads. For instance, it features a key labelled "envoi" (used to validate the instruction just keyed in) which has no counterpart on the TV controls. The fact that these options were taken almost without discussion is not surprising when we know that CA unit designers had previously worked on developing the Minitel. At each stage of design, they set the ground rules governing interactions between the user and his terminal and thereby incorporated in the system a particular concept of that user's aptitudes, frames of reference, and behaviour patterns.[3]

Summary

Without aiming at an exhaustive account, we have drawn up an inventory of methods employed by system designers to develop, promote, and impose a particular representation of the user. We have been concerned to show that despite apparent differences each method aims at articulating and developing user representations. Not all methods reviewed are of equal force. Contrary to what is expected and put forward by designers, implicit methods seem to be more powerful than explicit ones. As well, different methods correspond to different problems and circumstances. We particularly wish to stress the point that, even if our use of the term "user" tends to suggest the contrary, none of the constructed representations is merely the representation of

a particular facet which the "complete user" may display in a given situation. From this standpoint, the non-discriminating quality of the word "user," which is also an obstacle to its scientific aptness, well illustrates the incomplete synthesis between all the different postures offered by the representation techniques deployed and, sometimes, materialized in technological options. The problem for successful design as well as for CTA is how to deal with this proliferation of "users," each of whom corresponds to a specific situation. At some point, if they do not merge into a whole, they must at least become coherent with each other.

The Articulation of User Representations

From Local to Global Rationality

Arbitration between different representations of the user which may lead to conflicting pressures, or to the maintenance of certain outmoded user representations throughout the development process, is a problem that innovators have to deal with all the time. It would be unwise to overrate the rationality of the choices made. Often they are rationalized afterwards, when the corporate management (or for that matter the sociologist) asks the project leaders to describe how the project evolved. Situations of severe disagreement, for instance, taking the issue outside the laboratory or company bounds, also impose greater coherency of explanation.

We should not, however, regard these *ad hoc* decisions as purely arbitrary. Each of them can usually be justified at the local level by the configuration at a given point in time, i.e. the positions of the actors involved, the material factors at issue, and the nature of the problem that brings the actors together. Still, local rationality cannot explain in every case why some demonstrably relevant system characteristics were ignored. We have to look at the accumulation of separate decisions which builds up to an outcome that takes its final shape only when the whole process is complete. There is no absolute technological determinism at work in design processes. Each new option will rely on decisions taken at an earlier stage, but in most design processes there is enough latitude to neglect (some) elements of earlier decisions.

For instance in the CA unit case, the cable TV users in Montpellier were surprised and disappointed because they could not use their videorecorders as previously. They encountered three kinds of constraints. First, connecting the CA unit, the TV receiver, and the videorecorder was by no means a straightforward matter. It sometimes involved switching the connections depending on whether a programme was recorded or was viewed after it had been recorded. Second, it was not possible to view a programme other than the recorded one at a given time. Third, preprogrammed recording (i.e. in the user's absence) entailed severe constraints. When we try to explain these apparent design flaws, we come up against a paradoxical situation. On the one hand, a multitude of decisions affecting separate features of the CA unit can explain the failures. These *ad hoc* decisions were fully justifiable in the specific situation. Only the reconstruction by the analyst (prompted perhaps by end-user complaints) imposes a coherent sequence which led to a disqualification of the videorecorder. On the other hand, it cannot be said that the design flaws and subsequent disqualification were at any time a deliberate intention of the designers, or could be easily foreseen.

This raises the problem of how to simulate a confrontation between different user representations in the design-development phase and, from there, how to devise compromise solutions which ensure their superposition. In the following we will show that several different strategies can be applied to combine and superimpose separate representations of the user. These stategies involve not only a reallocation of product-definition responsibilities but also a rigidification of the technical objects conducted on a very different time schedule.

The needed compromise of user representations entails the alignment of networks. When we designate the user as a "customer", "telephone subscriber", "town-dweller", "parent", or whatever, this immediately calls up a picture of a whole set of actors, objects, and networks of relationships binding them together. The term "telephone subscriber" brings with it objects called "telephones", whose keyboards are laid out in a particular way, "actors" like the Telecom administration personnel, bills made out in a certain way (fixed component and pro rata charges, etc.) and a definition of the contractual relationships and responsibilities involved. In the case of the CA unit, the "telephone subscriber" reference invokes all these features either directly or by inference. As is already put forward in actor-network theory and shown for many cases, a new technological system will succeed only when it is able to attract a whole universe: a network of socio-technical relationships has to be put together, persuaded, and enlisted. In the final analysis, verifying the viability of the proposed combination of user representations entails determining if a system is able to relate harmoniously to appropriate networks, and ensuring that the various implications of the proposition are not conflicting and do not introduce intolerable stresses or constraints. This brings us to the various strategies available for achieving network alignment.

Aligning the Socio-Technical Networks

Three different strategies were observed in the course of the case studies performed: 1) relying on the technological system itself, 2) delegating reconciliation of representations to intermediaries belonging to established socio-technical networks, and 3) "bridging" networks put in place concurrently with the system's development. They differed according to the basic means adopted for the purposes of reconciling the varied facets of the user.

Delegating the alignment task to the system means endowing it with a number of physical features that enable it to assume all the functions this involves. This was the strategy adopted, almost to the point of paroxysm, in the case of the CA unit. It could be said that the system's designers produced a marvellous machine capable of coping with extreme market segmentation by narrow differentiation of both products and subscriber characteristics. They relied on the Minitel example which seemed to sanction a form of technological determinism. They failed, however, to construct a supply structure capable of giving proper shape to this new type of audiovisual market. From the very outset, the project's stated objective was to set up a multi-media system (hi-fi sound, TV, data-transmission, telephone etc.). This was technically feasible only through the use of optical fibre. The system was interactive enabling the user to communicate with the network operator. It could accommodate a range of scales of charges (basic subscription plus a large number of options, as well as restricted-access viewing for services open only to selected user categories (e.g. programmes produced by pharmaceutical companies and reserved for medical practitioners). This complex configuration meant that the CA unit had to handle transactions on two levels, the technical (selection and reception of programmes by users as desired) and the contractual (effective protection of restricted access services, clear notification of financial commitment undertaken by subscribing to particular services). The designers thus had to make provision for a number of access keys (numerical code words) for general use, for admission to restricted-access or surcharged programmes, and for preventing unauthorised use of the unit.

The access key for surcharged programmes was not included in the original plan, but gradually emerged as the solution to a problem that the designers had to face: How to ensure that a user acknowledges his debt when making a financial commitment to the network operator? As a solution, the designers incorporated on the front of the CA unit an additional feature, a "surcharge display" which informs the user about the selected programme's tariff status. The user signifies the acceptance of the conditions displayed by pressing twice on a key labelled "envoi." This is the formal equivalent of a signature and ensures that the "viewer" and "customer" facets of the user coincide. The second facet covers the contract relation with the network operator.

Once this point had been dealt with, yet another problem arose: the customer is already defined by another procedure. The signing of the subscription order and the additional procedure brings no guarantee that this signatory is the *same* user as the one who pressed twice on the "envoi" key. The possible non-coincidence of these two actors may give rise to difficulties, for example, when subscribers contest the amounts charged to them. The designers thus added a "paying access key" that was supplied to users on request, and whose purpose was to guarantee the existence of a legally responsible person backing up the user who presses twice on the "envoi" key. This grants power of decision to duly authorized members of the user household, freely conferred by the contract signatory, with the latter remaining solely responsible in the eyes of the network and its operator. This is a particularly clear illustration of the way in which the strategy adopted by the unit's designers tends to achieve the coincidence of different representations of the user entity by incorporating appropriate features in the technological system itself.

The "range of products" approach is another way of delegating the task of reconciliation to the artefact. In the case just discussed, a single artefact capable of coping with very different situations, expectations and requirements had to be developed. An alternative strategy consists in determining coherent sets of user characteristics. For example, if the business user and the domestic user are not likely to converge, a successful approach might be to develop two similar but not identical products, each tailored specifically for the identified user type.

On the whole, strategies that delegate reconciliation functions to the system hardware are based on a "technological determinist" concept of the innovation process. They elevate the innovator to the status of a demi-god, capable of foreseeing and handling user reactions. The process of identifying user representations is fully concentrated in the design phase. This involves the risk (which, in France at least, has often materialized) of ending up with a kind of technological monster, extremely sophisticated but finally quite ineffectual because it is unable to attract the users for whom it was intended. This excessive sophistication is not due to a pure love of technology, blind to the mundane conditions of the real world, but to an undiscriminating determination that the system itself should cope with the whole infinite variety of possible situations it might have to address.

Reconciliation of divergent user representations can also be delegated to intermediaries, the second general strategy. Such intermediaries might be "outsiders" regarded as able to provide entry to established socio-technical networks. These networks are themselves identified with the circulation of specified items (or systems) between actors whose own identities are defined in terms of that process of circulation. If such networks are to be able to channel the new system towards its intended users, a mutual rapprochement must be effected between the users and the system. This is the task the intermediaries have to perform.

The Securiscan system used a strategy of this kind. The designers started with the idea of a high-tech product adaptable to individual user specifications and distributed through supermarkets. The designers soon realised that such a configuration assumed a variety of user representations which were unlikely to converge in a single individual. These representations were those of the gadget-lover, the do-it-yourself addict, the person preoccupied with security, the computer specialist, the supermarket customer, the middle-to-high income earner, and so on. At a fairly early stage the decision was taken to abandon the distribution through supermarkets and instead rely on conventional dealers who were better able to sound out their clientele. The dealers were to work with any interested parties to devise a system configuration corresponding to their individual requirements, keeping within the limits of the technical possibilities. In addition they installed the system, programmed its implementation, provided after-sale services, and encouraged users to extend their installations over time.

A strategy of this kind assumes a special design approach. It must allow and enable the intermediary, after appropriate coaching, to make the necessary adjustments between user and system. The Securiscan designers had to satisfy just as much the user wanting to run his heating system with optimum efficiency, for example, as the one who is afraid of burglars, or wants his lawn to be watered when he is away, or his shutters to be closed automatically as well as to meet any *combination* of these requirements. In response the final version of Securiscan is basically an open-ended system comprising three types of components:

- a central programming unit governing the overall installation set in place by the intermediary;
- end-applications equipment (e.g. switching mechanisms with control doors or taps);
- interface elements which transmit the central unit's instructions to the applications.

It was an interactive system. The applications equipment could feed back information (such as thermostat readings) to the central unit, which then made decisions. The above example shows that the strategy consists in appointing intermediaries to organize the demand side, ensuring the convergence of disparate user representations. This strategy requires that product design must not be the sole prerogative of the people engaged in the design-development phase proper, and that the distributors must be qualified to make any finishing touches required.

The last of the three possible strategies identified was creating a new network. When the Contact Ambiance telephone was being brought out, telephone sets as such were not recognized as retail items in the French market. They were supplied by the Telecom administration to all new subscribers and thus considered as a natural extension of the telephone network itself. The problem thus arose of how to bring about the existence of a "telephone buyer." Initially, the manufacturer thought that the mere fact that people bought products of all kinds was sufficient to ensure that they would also buy telephones. Accordingly the intention was to sell the Contact Ambiance telephones through the mass retail outlets. However, although the telephone was a familiar object, a set of problems emerged which could not be solved through this style of marketing. In supermarkets, for instance, the telephones were displayed "unescorted." How could potential customers be persuaded that there was any point in buying telephones when they were already supplied by the Telecom administration at a nominal charge and replaced at no cost whenever necessary? And how could they be convinced that the products on display were of high standard and fully compatible with on-going plans for expanding the telephone service, then in full swing? The strategy adopted by the manufacturer focused on two main points: securing approved status for the "Contact Ambiance" range from the Telecom administration; and creating a specialised sales network.

In this case we do not have a pure example of creating a specific network because features of the "intermediaries" strategy were also present. The manufacturer sought to superimpose the "telephone subscriber" and "telephone buyer" with the aid of an existing network: the customer relations agencies of France Télécom whose function was to manage subscriber accounts at the local level. The manufacturer formed these into a distribution network, training their personnel in sales techniques, and providing them with sales literature and display cabinets. Since France Télécom was not prepared to act as the exclusive sales agents of one manufacturer, competitors could use the agencies as well. This had the advantage of presenting a wide range of choice

and reinforcing the credibility of all, as well as of discouraging the creation of rival networks.

In this particular instance, the product was not significantly redefined by the actors downstream of the original designer. Some changes were made. For example, the instruction leaflets initially intended to help the user left alone with his new instrument, were rewritten to make room for the interaction between the untutored buyer and a sales agent with specialist knowledge in telecommunications. The convergence of the two facets of "subscriber" and "buyer" was achieved partly through combined action of the administrative and the commercial networks, and partly due to the increasing diversification of the products comprising the range, in terms both of their aesthetic qualities and their functions.

Although considered separately for convenience of description, the three strategies reviewed above are rarely encountered in their pure form as is indeed demonstrated by our third example. In general, we find a mix of strategies with varying emphasis. The idea that a novel artefact can by itself achieve convergence among the universe of user representations is no more than an engineer's fantasy. Equally, it would be vain to pretend that a salesman can save an artefact that is unable to sustain *any* of the relationships that it seeks to develop. Developing and incorporating user representations during the design stage is, therefore, not merely a matter of forming a realistic image of the potential user. It is above all a matter of forming a binding association between technological objectives and marketing strategy. In other words, achieving the convergence of all the different user representations involves gaining control over both the technical and the commercial sides, and acting on the interface between them.

Conclusion

The methods employed for bringing the end-user into the design process are varied. They all define ways of representing users, and each produces a stylized picture of the situation. The main problem for design, and CTA, is not so much multiplying the number of approaches and resulting user representations for these already abound. To strengthen the design process by incorporating a multiplicity of user representations, the main challenge is to coordinate the application of the various methods and reconcile their results.

It is therefore productive to devise assessment tools that can be used to draw up a chart for each phase of a project. The chart could show all

the states of the user representation, how they have been constructed, and whether they are made into "given" assumptions by technological options already applied. The purpose of this exercise is threefold: First, it could verify the coherency (and potential for consolidation) of all the varied user representations as the project goes along. Second, the strategy should continually monitor potential ways to achieve convergence among disparate states of the user representation. Finally, it could help build a strategy for further project development. To develop this part of reconciling different user representations a bit further, I want to emphasize the binding complementarity of the decisions regarding the technical content of an innovative product and those shaping its marketing strategies.

For CTA it is of special importance to find ways of ensuring that certain user representations – which would otherwise not be considered by the innovators and the entrepreneurs – are taken into account. Conceivably, the whole range of methods and resources already in use, including financial incentives and standards, could superimpose neglected user representations. In addition, public authorities should identify, create and/or use a number of mediators between innovators and end-users to redefine the demand and thus allow new user representations to be considered. To implement this it will be necessary to develop a sociology of marketing and distribution able to become an integral part of the design, development, and innovation process.

In a way the redefinition of demand has already been extensively dealt with by economists. They pose the problem by asking, what mechanisms would correct shortcomings in the market? But the way we view these mechanisms, and the place given to different actors in the debate, can be considerably transformed if we are conscious of the omnipresence of the users in the process of conception, and of the main resulting problem: how to make the different representations coherent, or, in other words, how to construct "interessement devices" (Callon, 1986, 1987) which permit articulation between the largely heterogeneous techno-economic networks.

Notes

1. Joint research project culminating in a report edited by Dominique Boullier (1989).
2. Meaning that the new system should be able to work off the wiring already installed on the premises.
3. Industrial standards also impose certain assumptions concerning user characteristics. Once they have been adopted, they leave system designers with limited scope for discussion.

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9 Technologies as Social Experiments. The Construction and Implementation of a High-Tech Waste Disposal Site[1]

Ralf Herbold

For several decades, the introduction of new technologies, such as biotechnology and nuclear technology, has been accompanied by wide-ranging discussions about the risks involved. This has recently led to a new self-perception of society as an experimental society. In this chapter, a case study from the field of waste management, I will argue that the implementation of new technologies in fact increasingly takes on the character of a social experiment.[2] In this field the characteristics of the classic laboratory experiment have been integrated into the planning process. Like the classic laboratory experiment, the social experiment is characterized by verifiable hypotheses, an organized research process, and the definition of an experimental situation.

Within an industrial context some authors have identified similar developments. These include non-linear learning (Meyers 1990) and innovation as a recursive process (Asdonk et al. 1991). Here, the implementation of new technologies is increasingly guided by the insight that the performance of new technological systems is only partly predictable. Consequently, feedback loops become a fundamental element of exchange between suppliers and customers. New technologies are thus products of a social learning process.

Some forms of experimental implementation may be opposed by society because of perceived risks, and such opposition cannot be neglected. Indeed, issues of safety relating to the social impacts of new technological systems cannot be mediated solely by assessing objective risks and isolating technical arrangements from technical systems (see Wynne, this volume). Instead, social negotiations about the per-

Phys 570H 1995-01

Managing Technology in Society

The approach of Constructive Technology Assessment

Edited by
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Universiteitsbibliotheek
Utrecht

PINTER
PUBLISHERS
LONDON AND NEW YORK
Distributed in the United States by St. Martin's Press

NATUUR- EN STERRENKUNDE
BIBLIOTHEEK

PINTER
An imprint of Cassell Publishers Limited
Wellington House, 125 Strand, London WC2R 0BB, England

First Published in 1995

© The editors and contributors, 1995

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Distributed exclusively in the USA and Canada by St. Martin's Press, Inc., Room 400, 175 Fifth Avenue, New York, NY10010, USA

British Library Cataloguing in Publication Data

A CIP catalogue record for this book is available from the British Library

ISBN 1 85567 339 8 (hb)
1 85567 340 1 (pb)

Library of Congress Cataloging-in-Publication Data

A CIP catalog record for this book is available from the Library of Congress

Typeset in Book Antiqua by Bob Kennedy, Enschede, The Netherlands
Printed and bound in Great Britain by Biddles Ltd., Guildford and King's Lynn

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